Regulatory compliance: Strict regulatory frameworks must be adhered to by health AI technologies. Healthcare providers and software developers must ensure that AI-powered solutions conform to regulatory standards in tandem with the changes in these standards.

The role of AI in drug discovery and development

AI is transforming drug discovery and development, revolutionising traditional methods and accelerating the process of bringing new drugs to market. Here are the key aspects highlighting the role of AI in this crucial domain.

Target identification and validation

Traditional approach: Identifying and validating potential drug targets involves extensive biological research.

AI contribution: AI algorithms can analyse vast biological data sets, including of genomic and proteomic data, to identify and prioritise potential drug targets. For example, machine learning models can predict the likelihood of a specific protein being a viable target for drug intervention based on its role in disease pathways.

Drug design and optimisation

Traditional approach: Designing and optimising drug compounds traditionally involves time-consuming trial and error processes.

AI contribution: AI-driven platforms, like generative adversarial networks (GANs) and deep learning models, can assist in designing novel drugs by predicting molecular structures with desired properties. Atomwise, for instance, employs AI for virtual screening of potential drug compounds, saving time and resources in the early stages of drug discovery.

Drug repurposing

Traditional approach: Discovering new uses for existing drugs typically requires extensive clinical testing.

AI contribution: AI can analyse large data sets, including clinical records and molecular information, to identify existing drugs that may be repurposed for new therapeutic purposes. This can significantly shorten the time and cost associated with bringing a drug to market. For example, BenevolentAI utilises AI to identify existing drugs that may be effective against different diseases.

Predicting drug-drug interactions and side effects

Traditional approach: Assessing potential drug interactions and side effects traditionally involves conducting extensive clinical trials.

AI contribution: AI algorithms can predict potential drug interactions and side effects by analysing biological and chemical data. This helps in optimising drug combinations and minimising adverse effects. IBM Watson for drug discovery is an example of a system that uses AI to analyse scientific literature and databases to predict potential interactions and side effects.

Clinical trial optimisation

Traditional approach: Recruiting suitable patients for clinical trials and optimising trial parameters is a complex and time-consuming process.

AI contribution: AI assists in patient recruitment by analysing diverse data sets to identify suitable candidates. Additionally, machine learning models can optimise clinical trial protocols, helping to streamline the process and reduce costs. Tempus, for instance, uses AI to optimise clinical trial design and execution.

Drug manufacturing and quality control

Traditional approach: Ensuring the quality and consistency of drug manufacturing involves extensive manual testing and quality control procedures.

AI contribution: AI is employed for process optimisation in drug manufacturing, ensuring quality control, and minimising variations in production. This leads to more consistent and cost-effective manufacturing processes.

AI has the potential to revolutionise healthcare by improving diagnostic accuracy, reducing the time taken for diagnosis, and enabling personalised treatment plans. As AI-based technologies continue to evolve, it is crucial to address ethical considerations, data privacy concerns, and ensure collaboration between AI systems and healthcare professionals for optimal patient care. END

References

- *https://insights.omnia-health.com/technology/ai-powered-growth-healthcare-early-trends-and-learnings*
- *https://research.aimultiple.com/healthcare-ai-use-cases/*
- *https://www.jax.org/personalized-medicine/precision-medicine-and-you/what-is-precision-medicine*
- *https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10302890/*

By: Dr Magesh Kasthuri and Dr Anand Nayyar

Dr Magesh Kasthuri is a senior distinguished member of the technical staff and principal consultant at Wipro Ltd. This article expresses his views and not that of Wipro.

Dr Anand Nayyar is a PhD in wireless sensor networks and swarm intelligence. He works at Duy Tan University, Vietnam, and loves to explore open source technologies, IoT, cloud computing, deep learning and cyber security.